BIOCHEMISTRY CUMULATIVE EXAM

January 17, 1998

1. You have partially purified an enzyme by ion-exchange chromatography followed by preparative isoelectric focusing. You find using polyacrylamide gel electrophoresis in the presence of sodium dodecyl sulfate that five contaminating proteins are still present along with the enzyme of interest.

   a. (25 points) Based on the available information, what one single procedure would you do next to improve the purity of your enzyme? Explain why you would choose this procedure instead of other possibilities.

   b. (15 points) Explain the principles by which ion-exchange chromatography works. What materials, supplies and equipment are needed?

   c. (15 points) Explain the principles by which isoelectric focusing works. What materials, supplies and equipment are needed?

   d. (10 points) What is the purpose of the sodium dodecyl sulfate in the polyacrylamide gel electrophoresis experiment?

   e. (20 points) You are asked to determine whether the enzyme contains any disulfide bonds. How would you proceed? Explain how and why your method will work.

   f. (15 points) You are asked to determine number and sizes of subunits that make up the pure enzyme. How would you do this? Explain how and why your method will work.